

FLARE — Base Papers

Early Detection of Low-Rate Flow-Table Overflow (LOFT) Attacks in Software-Defined Networks

The foundational literature this work is built on and positioned against.

Primary base papers

These two define the project: the **attack** we defend against, and the **closest existing defence** we improve on.

1. FloRa — the defence we extend (lead base paper)

A. Mudgal, A. Verma, M. Singh, K. S. Sahoo, E. Elmroth, and M. H. Bhuyan, “FloRa: Flow Table Low-Rate Overflow Reconnaissance and Detection in SDN,” *IEEE Transactions on Network and Service Management*, 2024.

DOI: 10.1109/TNSM.2024.3446178 · preprint arXiv:2410.19832

Relevance. The most recent and closest published defence against LOFT attacks: a machine-learning classifier over flow-table features that detects and evicts malicious flows. FLARE is positioned directly relative to FloRa — same threat, aiming to add a time-to-overflow forecast and a training-free, online detection path. Read this one in full first; it sets the bar and appears in the target journal.

2. The LOFT attack — the threat we address

J. Cao, M. Xu, Q. Li, K. Sun, Y. Yang, and J. Zheng, “Disrupting SDN via the Data Plane: A Low-Rate Flow Table Overflow Attack,” *Proc. SecureComm*, 2017, pp. 356–376.

Relevance. Introduced the LOFT attack: quietly filling a switch’s flow table with low-rate flows refreshed just before their idle timeout, so the table overflows using almost no traffic and evades volume-based defences. Defines the two-phase threat model (reconnaissance → slow installation) FLARE targets.

Supporting foundation

3. Flow-table capacity / timeout inference (basis for the “probing” phase)

J. Leng, Y. Zhou, J. Zhang, and C. Hu, “An Inference Attack Model for Flow Table Capacity and Usage... in SDN,” *arXiv:1504.03095*, 2015.

Why it matters: an attacker can remotely infer the idle timeout and table size — the reconnaissance FLARE tries to detect early.

4. Denial-of-service via OpenFlow rule installation

R. Kandai and M. Antikainen, “Denial-of-Service Attacks in OpenFlow SDN Networks,” *IFIP/IEEE IM*, 2015, pp. 1322–1326.

Why it matters: establishes flow-table exhaustion as a practical DoS vector in SDN.

5. Feasibility of attacking SDN

S. Shin and G. Gu, “Attacking Software-Defined Networks: A First Feasibility Study,” *ACM HotSDN*, 2013, pp. 165–166.

Why it matters: early demonstration that the SDN control/data-plane interaction is attackable.

6. Why the flow table is small and expensive (the root vulnerability)

A. R. Curtis et al., “DevoFlow: Scaling Flow Management for High-Performance Networks,” *ACM SIGCOMM*, 2011, pp. 254–265.

Why it matters: documents the limited TCAM/flow-table capacity that makes overflow possible.

7. Change-detection method behind FLARE’s core detector

E. S. Page, “Continuous Inspection Schemes,” *Biometrika*, vol. 41, pp. 100–115, 1954.

Why it matters: the CUSUM procedure FLARE uses to detect slow, persistent table growth.

Notes. FloRa (#1) is the direct predecessor and is published in the intended venue (IEEE TNSM), so any comparison and novelty claim must be made against FloRa’s actual method, not a simplified baseline. FloRa’s citation (authors, venue, DOI) is verified; for the supporting papers, confirm exact page numbers on IEEE Xplore / the ACM Digital Library before formal submission.